

A numerical study of the function-level convergence in Suzuki’s conjecture on Weil’s quadratic form: the normalization constant is $\|\Phi\|_{L^2}$

Denis Joubert

Independent researcher, Switzerland

August 30, 2026

Abstract

We report a numerical study of Conjecture 1.2 of Suzuki (*Weil’s quadratic form via the screw function*, arXiv:2606.09096), carried out in the semi-local framework of Connes–Consani. The L^2 -normalized ground state v_a of the truncated Weil form on $[-a, a]$, computed from primes alone, converges in L^2 to $\Phi/\|\Phi\|$, where Φ is the inverse Fourier transform of $\xi(1/2 + iz)$: the overlap is 0.99964 at $\mu := e^{2a} = 11$, with L^2 deficit $\approx (\text{sup-norm residual})^2/2$ at every scale tested. This identifies the normalization constant of the conjecture: $c_a \rightarrow \|\Phi\|_{L^2(\mathbb{R})} = 1.130932026\dots$. Uniform convergence — the statement of Conjecture 1.2 *as written* — is much slower and is still at the few-percent level at these scales: the maximum relative residual follows $\approx \frac{1}{3}e^{-2a}$ and concentrates in the zero-free band $[0, \gamma_1)$, while the *zeros* of $\widehat{v}_a$ converge superexponentially (Connes; Groskin). The identification extends parameter-free to the five real primitive Dirichlet characters of conductor ≤ 8 , with exact theta kernels, to $\leq 4 \times 10^{-4}$. For these pole-free forms we also find a strikingly linear depth law $-\ln \lambda_{\min} = s(\chi) \cdot \mu$, with s increasing with γ_1 and a clear parity signature. These are measurements with stated error bars, not theorems; all code and a full lab notebook are public.

1 Setting and conventions

We follow Suzuki’s normalization $\xi(s) = s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$ and write $\Xi(z) = \xi(\frac{1}{2} + iz)$, so that Ξ is real, even, with $\Xi(0) \approx 0.9943$. Test functions live on the window $[-a, a]$, $L := 2a$, $\mu := e^L$; the semi-local Weil quadratic form Q_W contains the pole, archimedean and prime terms with primes $p^k \leq \mu$ (Connes–Consani, arXiv:2106.01715, §2). Matrices are assembled in the even real Fourier basis of that paper, using its closed convolution table (Lemma 2.6) and the $\psi^\#$ decomposition; an overall factor $\frac{1}{2}$ in our $\sigma = \frac{1}{2}Q_W$ is irrelevant for eigenvectors. Quadratic forms are diagonalized in multiprecision (`mpmath`, 40–130 digits) with composite Gauss–Legendre quadrature whose nodes are Newton-refined in multiprecision. The ground state v_a is the $L^2(dx)$ -normalized eigenvector of the smallest eigenvalue $\lambda_{\min}$; its transform is $\widehat{v}_a(z) = \int v_a(x)e^{izx}dx$. Following the conjecture we form $c_a := \Xi(0)/\widehat{v}_a(0)$ and compare $c_a\widehat{v}_a$ with Ξ .

2 Result 1: two-speed convergence and the constant (ζ)

(a) **Uniform convergence is slow and infrared-limited.** The N -extrapolated sup-norm residual obeys

$$R(\mu) \approx \frac{1}{3}\mu^{-1} = \frac{1}{3}e^{-L} \quad (0.33/5.5 = 6.0\%, \ 0.33/11 = 3.0\%, \ 0.33/16 = 2.1\%),$$

and at every μ it concentrates in the zero-free band $[0, \gamma_1)$: the residual there is 30–40 $\times$ larger than between zeros. The Γ -bulge of Ξ below $\gamma_1 = 14.13$ is the slowest-learned feature — the spectral measure lives on the zeros, and off its support the constraint is soft. By contrast the *zeros* of $\widehat{v}_a$ converge superexponentially at the same cutoffs (Connes arXiv:2602.04022 §6; Groskin arXiv:2605.20224).

μ	$\lambda_{\min}$	sup-residual (IR, N -extrap.)	c_a ($z=0$)
3.5	3.3×10^{-10}	$\approx 11.3\%$	1.2173
5.5	4.8×10^{-20}	$\approx 6.6\%$	1.180
7.5	9.3×10^{-30}	$\approx 4.7\%$	1.1648
9.5	4.1×10^{-38}	$\approx 3.4\%$	1.1553
11	3.6×10^{-48}	3.0%	1.1533
16	8.0×10^{-68}	$\approx 2.2 \pm 0.4\%$	≈ 1.147

Table 1: Ground state of the ζ form. Sup-residual = $\max_{z < 13} |c_a \hat{v}_a - \Xi| / \max |\Xi|$, extrapolated in basis size N (residuals converge *from below* in N ; small bases flatter the test). At $\mu = 11$, $N = 47$, our $\lambda_{\min} = 3.58 \times 10^{-48}$ approaches the published Connes–Consani value 2.389×10^{-48} from above, as Galerkin must.

(b) L^2 convergence is fast and identifies the constant. Let $\Phi_c(u) = \sum_{n \geq 1} (2\pi^2 n^4 e^{9u/2} - 3\pi n^2 e^{5u/2}) e^{-\pi n^2 e^{2u}}$ be the classical Riemann theta kernel. We verify numerically $\int_{\mathbb{R}} \Phi_c(u) e^{izu} du = \frac{1}{2} \xi_{\text{cl}}(\frac{1}{2} + iz)$ (ratio 0.500 flat in z), so in Suzuki’s convention $\Xi(z) = \int \Phi e^{izu} du$ with $\Phi = 4\Phi_c$ and

$$\|\Phi\|_{L^2(\mathbb{R})} = 1.130932026 \dots$$

The measured overlap $\langle v_a, \Phi \rangle / \|\Phi\|$ is 0.99964 at $\mu = 11$: no mass escapes to high frequencies, i.e. $v_a \rightarrow \Phi / \|\Phi\|$ in L^2 , hence $c_a \rightarrow \|\Phi\|$. The L^2 deficit matches $(\text{sup-residual})^2 / 2$ (3.6×10^{-4} vs. 4.5×10^{-4} at $\mu = 11$). The pointwise estimator $c_a = \Xi(0) / \hat{v}_a(0)$ inherits the local infrared error ($1.131 \times 1.02 \approx 1.153$ at $\mu = 11$, and its drift fits $c_\infty + 0.32/\mu$ — the same e^{-L} law), while the projection estimator $\|\Phi\|^2 / \langle v_a, \Phi \rangle$ converges quadratically: 1.13134 already at $\mu = 11$, within 4×10^{-4} of $\|\Phi\|$ and 3×10^{-3} away from the numerological candidate $2/\sqrt{\pi} = 1.12838$, which is thereby excluded.

Conjecture 1.2 thus splits numerically into a fast L^2 statement — our reformulation, verified to 4×10^{-4} at $\mu = 11$ with its constant identified — and the uniform statement *as the conjecture is written*, which at these scales still carries a few-percent residual governed by the infrared bulge at rate $e^{-L}/3$. We stress the distinction: nothing here verifies (1.2) at 4×10^{-4} ; that figure belongs to the L^2 version only, and the infrared bulge will not yield before much larger μ (a 0.1% uniform residual would require $\mu \approx 330$).

3 Result 2: the identification holds across the Dirichlet family

For a real primitive character $\chi \bmod q$ the (pole-free) form is built the same way, with the archimedean term evaluated through a Frullani representation of $\text{Re } \psi(\frac{1}{4} + \frac{a}{2} + \frac{it}{2})$ closed on $[0, L]$ (validated against the ζ pipeline to ten digits; the finite- T archimedean artifact of the literature is structurally absent). The exact theta kernels

$$\Phi_\chi(u) = 2e^{u/2} \sum_{n \geq 1} \chi(n) e^{-\pi n^2 e^{2u}/q} \quad (\chi \text{ even}), \quad \Phi_\chi(u) = 2e^{3u/2} \sum_{n \geq 1} \chi(n) n e^{-\pi n^2 e^{2u}/q} \quad (\chi \text{ odd})$$

satisfy $\int \Phi_\chi e^{izu} du = \Lambda(\frac{1}{2} + iz, \chi)$ with ratio 1.000 flat in z . Projection estimates of $c(\chi)$ at $\mu = 16$:

χ	q	parity	γ_1	$c_{\text{proj}}(\mu=16)$	$\ \Phi_\chi\ $ (exact)	difference
χ_3	3	odd	8.04	0.51558*	0.515314044	2.7×10^{-4}
χ_4	4	odd	6.02	0.815977	0.815799088	1.8×10^{-4}
χ_5	5	even	6.65	0.787253	0.786984626	2.7×10^{-4}
χ_7	7	odd	4.48	1.876290	1.875696997	3.2×10^{-4}
χ_8	8	even	4.90	1.283030	1.282526197	4.0×10^{-4}

Table 2: Parameter-free confirmation of $c_\infty(\chi) = \|\Phi_\chi\|$ (*: at $\mu = 11$). The residual differences are the finite- μ deficit and shrink quadratically: for χ_4 , $4.05 \times 10^{-4} \rightarrow 1.78 \times 10^{-4}$ from $\mu = 11$ to 16, ratio 2.28 vs. $(16/11)^2 = 2.12$.

4 Result 3: a linear depth law with a parity signature

For the pole-free forms, $-\ln \lambda_{\min} = s(\chi) \cdot \mu$ with striking linearity (segments $5.5 \rightarrow 11 \rightarrow 16$; basis-converged, $\lambda_{\min}$ shifts $\leq 2.4\%$ under matched bases):

	χ_8	χ_7	χ_5	χ_4	χ_3	ζ
γ_1	4.90	4.48	6.65	6.02	8.04	14.13
s	$1.47 \pm .05$	$1.58 \pm .05$	$2.41 \pm .04$	$2.93 \pm .04$	$4.00 \pm .07$	≈ 10 (nonlinear)
shape const. C	0.53	0.43	0.50	0.39	0.41	0.33

The depth s grows with the width γ_1 of the zero-free infrared desert: the quasi-radical digs where a test function can concentrate its spectrum unconstrained, and the pole of ζ acts by pushing γ_1 to 14.13 (its ladder, $\approx 10\mu$ but nonlinear over our range with matched bases, is consistent with the Connes–Consani asymptotics). Parity marks both observables in opposite directions: odd characters dive faster than even ones at comparable γ_1 ($\chi_4 > \chi_5$ and $\chi_7 > \chi_8$, two concordant inversions), while even characters carry the larger constant C of the shape law $R \approx C e^{-L}$ ($C \approx 0.50$ – 0.53 even, 0.39 – 0.43 odd, 0.33 for ζ). The naive candidate $s = \gamma_1^2/(2\pi e)$ is falsified by χ_4 (2.93 measured vs. 2.12; robust across three cutoffs). The exact form of $s(\gamma_1, \text{parity})$ is open. We note that this baseline is the prerequisite for a “Landau–Siegel seismograph”: an exceptional zero contributes a real, pole-like term to the explicit formula, so a real even character whose ladder is anomalously deep *for its* (γ_1, parity) would be flagged.

5 Methods, validation anchors, artifact taxonomy

Every pipeline is validated against independent anchors: the pole term against the closed form $32 \sinh^2(L/4)/L$; the archimedean term against its spectral definition $Q_\infty = \int |\hat{f}|^2 \frac{2\theta'(t)}{2\pi} dt$ and against the coefficient 2.00963 of Connes–Consani (their Fig. 4 value at $L = \log 2$); full matrices against the zero side of the explicit formula (280 zeros of ζ , 40–70 zeros per $L(s, \chi)$, computed independently; Λ real on the critical line to 10^{-31}); and $\lambda_{\min}(\mu=11)$ against the published 2.389×10^{-48} . Six artifact families were identified and neutralized along the way, and are documented in the repository: float64 eigenvalues producing fake positivity violations below $10^{-15} \|K\|$; sieve truncation creating off-line pseudo-zeros; the finite archimedean cutoff T (Groskin’s diagnosis; absent here by construction); float64 quadrature *nodes* flooring entire matrices at 10^{-16} ; splitting smooth integrands into near-divergent halves; and the two-limit protocol (shape residuals converge from below in basis size, so one must extrapolate in N before fitting in μ).

6 Discussion

The contrast between superexponential zero convergence and e^{-L} shape convergence is, we believe, the informative fact: the quadratic form pins its ground state on the zeros with prolate-scale violence, but learns the shape of Ξ between them only at one e -fold per unit of window — and slowest of all in the zero-free infrared. Open questions we would value expert views on: (i) is the L^2 formulation $v_a \rightarrow \Phi/\|\Phi\|$, with constant $\|\Phi\|$, already known or provable by current methods? (ii) what is $s(\gamma_1, \text{parity})$? (iii) does the generic pre-plunge closure rate of the band-limited form (≈ 0.41 per degree of freedom, measured robustly) have a prolate-theoretic derivation?

Verification status

Two tiers should be distinguished. *Re-derivable directly by the reader* (one-line quadratures of the explicit theta kernels, no pipeline needed): the Fourier identity $\widehat{\Phi}_c = \frac{1}{2} \xi_{\text{cl}}(\frac{1}{2} + iz)$ (ratio 0.500, flat in z), the factor $\Phi = 4\Phi_c$, the norm $\|\Phi\|_{L^2} = 1.130932026\dots$, the five Dirichlet norms $\|\Phi_\chi\|$ of Table 1, and the internal consistency of the conventions once ξ_{cl} is fixed. *Author’s measurements* (public pipeline, multiprecision, replayable from the repository but not re-derivable by hand): the overlap 0.99964, the projection estimators c_{proj} of Table 1, and the depth slopes $s(\chi)$. Independent replay of the second tier is invited and takes minutes per entry.

Reproducibility

All code (mpmath/numpy only), zero caches, the full lab notebook, and per-figure commands with runtimes:

<https://github.com/darreal44/riemann-weil-phenomenology>

Acknowledgments

Computations and analysis were carried out with the assistance of Claude (Anthropic). All claims, and all remaining errors, are the author's.

References

- [1] M. Suzuki, *Weil's quadratic form via the screw function*, arXiv:2606.09096 (2026).
- [2] A. Connes, C. Consani, *Spectral triples and ζ -cycles*, L'Enseign. Math. 69 (2023); arXiv:2106.01715.
- [3] A. Connes, C. Consani, H. Moscovici, *Zeta spectral triples*, arXiv:2511.22755 (2025).
- [4] A. Connes, W. van Suijlekom, *Quadratic forms, real zeros and echoes of the spectral action*, arXiv:2511.23257 (2025).
- [5] A. Connes, *The Riemann hypothesis: past, present and a letter through time*, arXiv:2602.04022 (2026).
- [6] A. Groskin, *High-precision approximation of Riemann zeros via the truncated Weil form*, arXiv:2605.20224 (2026).
- [7] D. Slepian, H. Pollak, *Prolate spheroidal wave functions...*, Bell Syst. Tech. J. 40 (1961).